

Sponsor Meeting Notes: October 1, 2026

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Meeting details

Field	Value
Date	October 1, 2026, about 3:15 PM
Length	About 44 minutes
Attendees	Dr. Qualls (sponsor) and the student team (Joe and Chris heard on the recording)
Absent	Craig, Dr. Qualls's former PhD student and author of the original code, who answered the question list by email
Purpose	Walk through the team's question list and hand over the existing code

Materials delivered

Dr. Qualls answered the team's question list line by line, then sent the same questions to Craig and appended his separate answers. Both sets are in the shared OneDrive folder.

- **Response to Team Query (Word):** both sets of answers, Dr. Qualls's first and Craig's below.
- **FDL Processing Script Information (Word):** Craig's description of what the FDL model does, including how it chooses a starting point. Dated, so later versions can be told apart.
- **The FDL Python script:** uploaded as a .txt file because Craig could not send .py. Rename it to run it.
- **Woodruff, Qualls and Humes paper:** the threshold analysis, directly relevant to this project and not heavy on math.
- **Woodruff, Qualls and Clark paper:** published earlier this year on the Boise basin, three watersheds north of Boise, framed around drought.

Craig is out until Monday. He will then send the Snake watershed boundary or mask, the output FDL rasters, the PC1 rasters, and the runtime settings he used. Dr. Qualls is checking how much OneDrive capacity he has in case the data is too large.

Threshold: resolved

Default the tool to 10 for building the FDLs and the PCA, with an advanced option to change it.

- The 2019 paper used about 40, which is the usual value in the literature for deciding snow or no snow in a single image.
- The Woodruff, Qualls and Humes paper tested 10, 15, 20, 25 and 40 and found that **10 produces a crisper, more robust PCA**. Dr. Qualls suspects this is because a lower threshold is reached later in the melt season, when there is less cloud cover than in winter. That explanation is unverified.
- The Boise basin paper may have used 10, but he was not certain.

The gap-fill threshold is independent

This is the part worth building around. The threshold used to cloud gap fill a given day does not have to match the threshold the PCA was built with.

- The melt pattern stays essentially the same for any threshold up to about 50, so a PCA built at 10 can gap fill an image at 15, 30, 40 or 38.5.
- That means a user can pick any snow line they want without recomputing anything.
- Taken further: run the gap fill at 30, then 29, 28, 27 and so on, and reassemble the results into a single image with nested snow lines. That approximates a full NDSI image rather than a binary snow map.

Data product: use the cloud gap filled files

This changes what the team has been pulling. Craig's script uses NASA's **cloud gap filled** product (the 10A1F files), not the raw daily NDSI images.

- In the raw images, a cloudy pixel simply reports cloud, so nothing is known about the ground.
- In the cloud gap filled product, NASA looks back to the last time the pixel was seen as snow or land and reports that value. If a pixel was snow and then sat under cloud for 14 days, it reports snow for all 14 days.

- The metadata records **how many days ago that observation was**, so inferred values are already flagged. That answers how the tool should label inferred pixels for the end user.
- Dr. Qualls thinks it is easier for the FDL code to work with this product than to handle cloud values itself.
- One thing to check in the script: when it jumps into a date whose value is five days stale, does it step back to the day the snow was actually last seen? He would guess yes, but was not sure.

He also noted that the 2019 paper predates this product and did not use it, and that he believes the 2019 paper influenced the way NASA built it. Dorothy Hall is the NASA investigator responsible for the MODIS snow algorithms and what gets archived at NSIDC.

How the FDL step works, and why clouds wash out

The search. The script jumps into a date sitting in the middle of a long stretch of continuous snow cover (March 30 was the example), then walks forward through subsequent images, pixel by pixel, until land first appears. That date is marked as the first day of land. Starting inside a continuous snow period is what keeps a transient snowfall after an early melt-out from corrupting the result. Craig has tuned this over several years, and the chosen start date is specific to the Northwest and similar melt-dominated northern watersheds.

Why clouds do not ruin it. A cloudy stretch pushes the recorded date to the end of the gap for first day of land, or to the beginning for last day of snow. That enters the FDL as noise. Because clouds are random from year to year, the noise does not repeat, so the first principal component keeps only what the years have in common.

The evidence. Craig computed the PCA from first day of land and again from last day of snow. The two bias in opposite directions, so any surviving cloud contamination would show up as a difference between them. The correlation came out around 0.99999, which is why the choice between first day of land and last day of snow is arbitrary. First day of land was picked because the question is about melt-out.

If results look odd. Dr. Qualls invited the team to set Craig's approach aside and reason from scratch about how they would find the start point given the same data. An algorithm that is faster or more accurate would be welcome.

Architecture: confirmed

Precompute once. For a given threshold, the FDLs are computed once and stored forever. Each new year adds one FDL image. This is by far the most compute-intensive step.

FDL ignores watershed boundaries; PCA does not. The FDL step can run across whole tiles. Craig once processed the entire Northwest at once. The stored FDLs are then clipped with a shape file and the PCA is run per watershed.

Why PCA must stay local. Snowpack varies independently between regions: a high year in the Boise basin may coincide with a medium year in northern Idaho and a low year in the Washington Cascades, and those relationships flip from year to year. Combining them confounds the PCA. Within a limited watershed, the spatial pattern of melt stays consistent even between high and low snow years.

Speed. For the upper Snake, producing the PCA from stored FDLs took about five seconds.

Year selection. Because FDLs are stored per year, an advanced user could choose which years to include, for example only drought years, only non-drought years, or a mix.

Implementation details

- Craig used scikit-learn's PCA, which is a singular value decomposition, with the FDL matrices fed in directly.
- Centering and rescaling matter. A year where melt takes 150 days and a year where it takes 75 must be centered and rescaled to span the same period, not overlaid raw. The library does this, but it has to be doing it. Rescaling is what removes the effect of a particular year being a high snowpack year or having an early start.
- Covariance versus correlation was not settled. Dr. Qualls thinks covariance gives more weight to data that took longer to melt, but said not to quote him. Craig's answer pointed to the library rather than stating it explicitly.
- He contrasted this with other published approaches that average a given calendar day across years to get a melt-out probability. Those are tied to a historical average rather than the actual year being corrected. Centering and rescaling is what avoids that problem.

Cloud gap fill, in code terms

Craig's gap fill routine scans the image for places where a snow pixel sits next to a no-snow pixel, which marks the visible snow line. It then finds the contour in the PCA image that runs along that line, and traces that contour around the watershed: snow above, no snow below. The PCA image is a non-dimensional scale of each pixel's melt timing relative to the others, so a contour is simply a line of equal melt timing. No off-the-shelf library does this; Craig wrote it. He is still updating two stages, FDL to PCA and PCA to cloud removal.

Hosting, language, and support

Hosting. Dr. Qualls is working with Research Computing and Data Services at U of I. The Idaho Water Resources Research Institute, which his project is associated with, has already spoken with them and they are interested in helping. No commitments yet, but they can at least point to where this gets hosted. Near-term storage needs are small enough not to block work.

Language. Python is fine. C or another compiled language is acceptable if it runs the FDL step faster, as long as the result can be deployed publicly as an executable with no license requirement.

Support. Craig can answer implementation questions more thoroughly than Dr. Qualls can. He comes from hydrology, GIS and R rather than computer science, so Dr. Qualls expects useful exchange in both directions on how to streamline the processing. Dr. Qualls is available throughout the semester and wants regular contact.

Stretch goals Dr. Qualls would like to see

He was explicit that these are beyond scope and not expected, but said they would be great if time allows.

- **Find the start date automatically.** Instead of being handed a date, have the algorithm search a year of data for the longest continuous period of full snow cover and jump in there. He floated using AI to identify that period.
- **Make it hemisphere-agnostic.** Test that generalization on a watershed in the Andes, where midwinter falls at the end of June, or in the Himalayas, where accumulation may follow the monsoon rather than the northern winter. Idaho watersheds alone will also vary: a high-elevation basin may want a different date than a low one.

- **Expand geographically.** He wants this extended to the West Coast and eventually worldwide, picking watersheds in the Andes, the Himalayas and the Alps.
- **The long-term goal.** Get the National Snow and Ice Data Center to adopt this as a standard cloud gap filled product they archive and distribute, at least as an alternative to the current one.

Correction: USGS watershed boundary codes

Dr. Qualls recalled that USGS publishes nested watershed boundaries at several scales but could not remember the term and had the numbering backwards, saying larger numbers mean larger areas.

The term is the **hydrologic unit code (HUC)**, from the USGS Watershed Boundary Dataset, and **more digits means a smaller area**:

Code	Level	Example scale
2 digit	Region	The Columbia basin
4 digit	Subregion	
6 digit	Basin	
8 digit	Subbasin	
10 digit	Watershed	The watersheds above Boise
12 digit	Subwatershed	

This matters for the tool because PCA quality depends on keeping the area limited. The team should decide which level the watershed selector offers, probably 8, 10 or 12 digit units, and confirm with Dr. Qualls and Craig that the choice matches what the method was validated on.

Action items

Owner	Task	Due
Craig	Send the Snake watershed boundary or mask, the output FDL rasters, the PC1 rasters, and the runtime settings used	Monday
Dr. Qualls	Check OneDrive capacity against the volume of data Craig will send	Before Monday
Dr. Qualls	Continue with Research Computing and Data Services on hosting and storage	Ongoing
Team	Rename the .txt script to .py and read it alongside the FDL processing description	Now
Team	Pull a melt season of cloud gap filled data and run the FDL script on it to see whether a usable FDL comes out	Now
Team	Read the Woodruff, Qualls and Humes threshold paper, and the Boise basin drought paper	Now
Team	Verify in the script whether a stale cloud gap filled value is walked back to the last clear observation	Now
Team	Decide which HUC level the watershed selector uses and confirm it with the sponsor	Open
Team	Prepare implementation questions for Craig, who can answer the finer details	Before contacting him